

Research and Adaptation of Components from the Persistent OS Phantom for Genode

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Data Persistence

Data persistence: period of time during which data exists and is used

R. Morrison, M. P. Atkinson — “Persistent Languages and Architectures”, 1990

Today's dominant model — **two-level store**:

- ▶ Data is either *active* (RAM) or *stored* (disk)
- ▶ Programmer manually moves data between levels
- ▶ Consequences: data loss, I/O boilerplate ($\approx 30\%$ of code)

Orthogonal Persistence

Alternative: the **environment** manages persistence

Three principles:

1. **Independence** — data persistence does not depend on how it is manipulated
2. **Data type orthogonality** — all types can be persisted
3. **Identification** — type does not affect persistence identity

Benefits: up to 30% reduction in code size, more efficient use of disk bandwidth and improved reliability fault tolerance.

PhantomOS

PhantomOS is a non-Unix like OS implementing orthogonal persistence (D. Zavalishin).

- ▶ Persistent Virtual Machine (PVM): bytecode VM + POSIX layer
- ▶ Snapshots saved periodically via **Copy-on-Write**
- ▶ Only changed pages written to disk (incremental)
- ▶ Programs paused only briefly during snapshot preparation

Currently ported to the **Genode** OS framework:

- ▶ Proven microkernels (NOVA, seL4, Fiasco.OC)
- ▶ Capability-based security, minimal TCB

Current snapshot mechanism: **Snapper**

Snapper

File-system-based generational snapshot storage.

Previous approach (“superblock”): snapshots stored as monolithic objects.

- + Fast
- No fault tolerance, no storage recovery

Snapper 2.0:

- ▶ Unchanged pages **linked**, not copied across generations
- ▶ ext4 + fsck for crash recovery
- ▶ Integrity checks (hashes per file)
- ▶ Configurable redundancy policy

Problem Statement

PhantomOS snapshots capture the **entire machine state** — including sensitive data present in memory.

Snapshots are stored in **plaintext**. An attacker with physical access can mount and read them directly, or fully reproduce the machine state.

- ▶ No prior research on this for PhantomOS
- ▶ No ready-made solution exists in Genode

Goal: implement a mechanism for the secure storage of snapshots.

Use Cases & Requirements

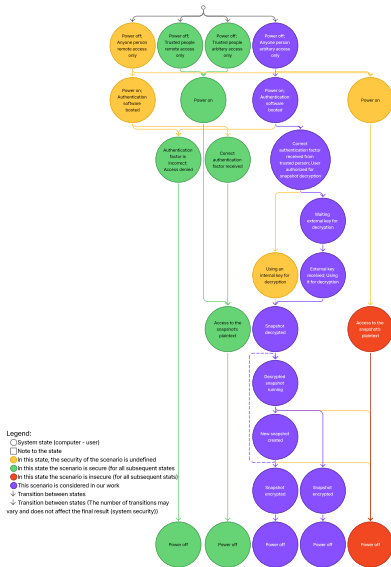
Snapshot must remain **encrypted** at all times.

Requirements:

1. Mandatory authorization
2. Key stored on an **external USB drive**
3. Encryption of snapshots (AES, block-level)
4. Only encrypted data written to disk
5. Acceptable performance overhead ($\leq 2\times$)

Requirements 1 & 2 combined: the USB drive contains the key and serves as the authentication factor.

Use Cases: Supported Scenarios



Implementation: `se_vault`

Genode includes **file_vault** + **Tresor** library:

- ▶ Tresor: AES encryption on 4K blocks, SHA-256 integrity, CoW atomicity
- ▶ `file_vault`: state orchestrator providing a secure FS session

Adapted `file_vault` → **`se_vault`**:

- ▶ GUI removed — headless operation
- ▶ Passphrase replaced by **USB key file**
- ▶ `rump_fs/ext2` replaced by **`lwext4/ext4`**
- ▶ Added **`fsck`** crash recovery support
- ▶ Direct block session (skip image file creation)
- ▶ All block sizes standardized to 4K
- ▶ Graceful shutdown sequence added

Related Fixes & Adaptations

lwext4 bug fixes:

- ▶ `ftruncate`: pointer cast for correct file size
- ▶ `sync` call now propagated to lower VFS layers
(critical — Tresor updates its state hash on `sync`)

Shutdown sequence (isomem / snapper / VFS):

- ▶ isomem: power-off button triggers final snapshot
- ▶ snapper, VFS: client counters trigger graceful shutdown
- ▶ lwext4 unmounted cleanly; mount flag cleared for `fsck`

Optimization: skipping image file pre-allocation
reduced initialization time from ≈ 20 min to ≈ 1 min.

Testing

Environment: QEMU with `-cpu host` (enables AES/SHA hardware acceleration).

2 GB image for isomem, 5 GB image for se_vault (4 GB user FS).

Functional tests (all passed):

- ▶ Normal boot / shutdown / restore cycle
- ▶ Encrypted image cannot be mounted without key
- ▶ Key/hash replacement attack — superblock not detected
- ▶ Crash recovery via fsck

Performance: first snapshot creation and restore, 5 runs each.

Results

Operation	Without enc.	With enc.	Overhead
Snapshot creation	2.10 s	51.00 s	$\approx 24\times$
Snapshot restore	2.50 s	27.30 s	$\approx 11\times$

Root cause: **known Tresor library limitation** — not optimized for multithreading. Acknowledged by the author; unresolved.

Default QEMU `-cpu Nehalem-v2` disables AES/SHA hardware acceleration; switching to `-cpu host` improved results significantly, but overhead remains high.

Evaluation

4 of 5 requirements met.

- ✓ Mandatory authorization (USB key)
- ✓ Key on external medium
- ✓ AES block-level encryption
- ✓ Only encrypted data on disk
- × Overhead $\leq 2\times$ (*Tresor limitation*)

Security gain: snapshot cannot be mounted or read without the key.

Integrity: SHA-256 at block level (Tresor) + hash at FS level (Snapper).

Conclusion

Adapted **file_vault** → **se_vault** for encrypted snapshot storage in PhantomOS on Genode.

First security research specifically for PhantomOS.
No analogous component existed prior to se_vault.

Future work:

- ▶ Fix Tresor performance (or switch to FS-level encryption via OpenSSL)
- ▶ Encrypt the isomem block device using se_vault
- ▶ Investigate merging Snapper and Tresor snapshot concepts